

Costs and Benefits of a Fossil Fuel Moratorium on Electrical Cooperatives and Municipal Boards in Virginia

Environmental Solutions Initiative Rapid Response Group

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Overview

Summary

- Nearly 1 in 4 Virginia customers (Table 1.1) receives electricity from outside the investor-owned utilities (IOUs). The other providers include 15 municipal boards (Table 1.2), the Virginia Tech campus, 13 cooperatives (Table 1.3), and third-party sources.
- Many providers operate in rural territories (e.g. Figure 1.1), leading to slightly higher electricity rates on average (Table 1.1).
- Nine of the 13 co-ops are part of the Old Dominion Electric Cooperative (ODEC), a coalition that owns several natural gas and emergency diesel plants, as well as partial stakes in a coal and nuclear plant (Table 1.4).
- Otherwise, most providers have little-to-no generation capacity of their own. Instead, they purchase some or all of their electricity from the grid or other suppliers, sometimes through multi-year contracts to reduce price volatility.
- On the supply side, several merchant-owned companies operate in-state power plants that sell directly to the grid. These outside parties are responsible for most of the recent and proposed additions to Virginia's fossil fuel portfolio (Table 1.5).

Notable Policies

- The Virginia Clean Economy Act (VCEA) includes a temporary moratorium on new fossil fuel power plants (to 1/1/22), and requires that any future proposals be scrutinized with respect to need and environmental trade-offs (including the social cost of carbon). However, the moratorium only extends to IOUs.
 - The VCEA's 2050 goal of 100% clean electricity *does* apply to all producers in VA
- Also, Virginia's 2021 entry into the Regional Greenhouse Gas Initiative (RGGI) includes **all** power plants with > 25 MW capacity. This includes plants owned by IOUs, ODEC, and third-party generators (Figure 1.3).
- Since Virginia is part of the 13-state PJM grid, leakage is a major concern. Policies like VCEA, RGGI, or a moratorium could shift third-party power plant proposals to neighboring states that lack those restrictions.
 - On the other hand, PJM [may not need](#) additional capacity for now. Since 2002, PJM peak demand has grown by 1% while total generating capacity is up 173%.

Table 1.1: Summary of electricity providers, customers, and sales in Virginia

	Full service providers				Other providers		
Item	Investor owned	Public	Cooperative	Non-utility	Energy	Delivery	Total
Number of Entities	4	16	13	1	6	3	
Number of Retail Customers	3,066,812	168,878	666,789	1	96	NA	3,902,576
Percentage of Retail Sales	81.2	3.6	13.8	0	1.4	NA	100
Average Retail Price (cents/kWh)	9.3	10.36	10.96	7.25	5.31	0.38	9.52

Source: Table 9 of EIA Virginia summary, based on data from EIA Form 861.

Table 1.2: Customers, Generation, and Electricity Sources of 16 Municipal Boards in Virginia

Power Provider	Residential Customers	Commercial Customers	Industrial Customers	Generation (MWh)	Purchases	Energy Source
Town of Bedford	5,780	886	28	14,536	206,761	5 MW hydro, purchases
Town of Blackstone - (VA)	1,749	281	5	13,298	24,696	purchases
Bristol Virginia Utilities	13,618	2,723	4	0	473,707	purchases
Town of Culpeper - (VA)	4,679	886	16	293	111,228	purchases
City of Danville - (VA)	36,850	10,956	16	35,289	944,409	11.1 MW hydro, 5.4 MW petroleum, purchases
Town of Elkton	929	156	6	0	21,926	purchases
City of Franklin - (VA)	4,626	842	0	90	139,121	6.4 MW petroleum, purchases
Town of Front Royal	6,656	999	.	0	190,448	purchases
City of Harrisonburg - (VA)	17,843	3,496	14	865	750,314	44.5 MW petroleum, purchases
City of Manassas - (VA)	13,828	2,664	1	847	425,815	54.8 MW petroleum, purchases
City of Martinsville - (VA)	6,539	1,121	5	6,446	172,665	1.3 MW hydro, 1.1 MW landfill gas, purchases
City of Radford -	6,579	861	32	6,394	177,466	1.0 MW hydro, purchases

(VA)						
Town of Richlands - (VA)	2,045	473	.	0	59,093	purchases
City of Salem - (VA)	11,270	2,184	78	0	379,717	5.1 MW petroleum, purchases
Town of Wakefield - (VA)	423	107	.	0	13,018	purchases
VA Tech Electric Service	5,476	1,143	5	0	348,346	6.3 MW natural gas & CHP plant, purchases

Source: EIA-861 (customers, generation, purchases), EIA-860 (energy sources)

Table 1.3: Customers, Generation, and Energy Sources of 13 Electricity Cooperatives in Virginia

Power Provider	Residential Customers	Commercial Customers	Industrial Customers	Net Generation (MWh)	Wholesale Power Purchased (MWh)	Energy Source
A & N Electric Coop	31,427	4,183	8	596	743,457	ODEC; 2.4 MW of petroleum
BARC Electric Coop	12,010	790	.	778	193,046	ODEC
Central Virginia Electric Coop	33,334	4,124	26	0	796,626	Purchases, some community solar
Community Electric Coop	9,334	1,868	2	0	209,350	ODEC
Craig-Botetourt Electric Coop	5,908	838	.	0	92,775	Purchases
Mecklenburg Electric Cooperative	29,385	1,814	28	0	630,201	ODEC
Northern Virginia Elec Coop	159,993	13,302	153	0	6,055,878	PJM purchases ; own biomass and renewables
Northern Neck Elec Coop, Inc	16,750	2,461	.	0	292,300	ODEC
Powell Valley Electric Coop	6,547	1,659	1	0	577,587	Tennessee Valley Authority
Prince George Electric Coop	10,517	1,365	16	0	359,868	ODEC
Shenandoah Valley Elec Coop	80,398	15,064	93	0	2,540,408	ODEC
Southside Electric Coop, Inc	54,428	2,375	17	0	1,005,891	ODEC
Rappahannock Electric Coop	154,119	11,899	553	0	4,060,522	ODEC

Source: EIA-861 (customers, generation, purchases), EIA-860 and websites (energy sources)

Figure 1.1: Map of Co-op Territories

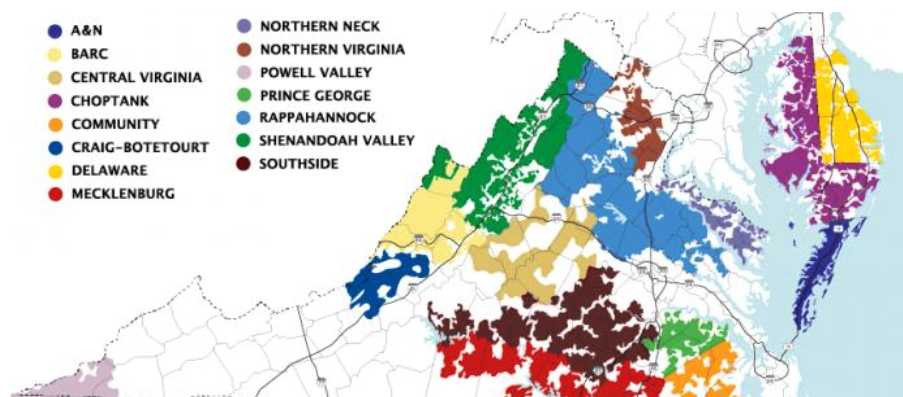
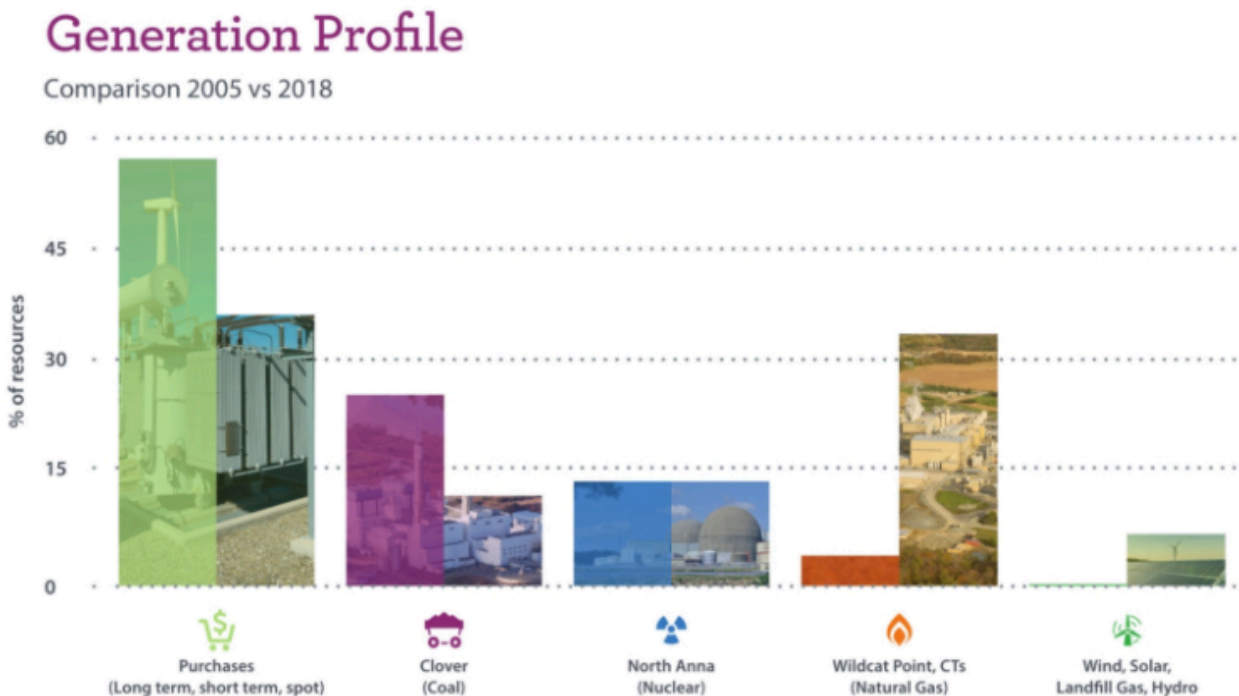


Table 1.4: ODEC-Owned Generation Facilities

Generating Facility	Ownership Interest	Location	Primary Fuel	Commercial Operation Date	Net Capacity Entitlement ⁽¹⁾
Wildcat Point	100%	Cecil County, Maryland	Natural Gas	04/2018	973 MW
North Anna	11.6%	Louisa County, Virginia	Nuclear	Unit 1 – 06/1978 ⁽²⁾ Unit 2 – 12/1980 ⁽²⁾	110 MW 110 MW 220 MW
Clover	50%	Halifax County, Virginia	Coal	Unit 1 – 10/1995 Unit 2 – 03/1996	220 MW 218 MW 438 MW
Louisa	100%	Louisa County, Virginia	Natural Gas ⁽³⁾	06/2003	504 MW
Marsh Run	100%	Fauquier County, Virginia	Natural Gas ⁽³⁾	09/2004	504 MW
Distributed Generation	100%	Multiple	Diesel	07/2002 05/2016	20 MW 6 MW 26 MW
Total					2,665 MW

Source: ODEC [Form 10-K, Year 2019](#)

Figure 1.2: Fuel Mix of ODEC Electricity Supply (Including Purchases)



Source: [2019 Report on Sustainability and Environmental Stewardship](#)

Table 1.5a: Recent Fossil Fuel Power Plants in Virginia (2017-2019)

Utility Name	Plant Name	County	Technology	Name-plate Capacity (MW)	Operating Month	Operating Year
Virginia Electric & Power Co	Greensville County Power Station	Greensville	Natural Gas Fired Combined Cycle	663.9	12	2018
Virginia Electric & Power Co	Greensville County Power Station	Greensville	Natural Gas Fired Combined Cycle	369.8	12	2018
Virginia Electric & Power Co	Greensville County Power Station	Greensville	Natural Gas Fired Combined Cycle	369.8	12	2018
Virginia Electric & Power Co	Greensville County Power Station	Greensville	Natural Gas Fired Combined Cycle	369.8	12	2018
Green Energy Partners LLC	Stonewall	Loudoun	Natural Gas Fired Combined Cycle	338	4	2017
Green Energy Partners LLC	Stonewall	Loudoun	Natural Gas Fired Combined Cycle	237	4	2017
Green Energy Partners LLC	Stonewall	Loudoun	Natural Gas Fired Combined Cycle	237	4	2017
Doswell Ltd Partnership	Doswell Energy Center	Hanover	Natural Gas Fired Combustion Turbine	187	3	2018
Doswell Ltd Partnership	Doswell Energy Center	Hanover	Natural Gas Fired Combustion Turbine	187	4	2018

Source: EIA-860 (2019), Table 3

Note: Virginia Electric & Power Co is now known as the IOU Dominion. All other utilities in Table 5 are third-party, merchant-owned generators. Does not include ODEC's Wildcat Point natural gas plant (Table 4), which became operational in April 2018.

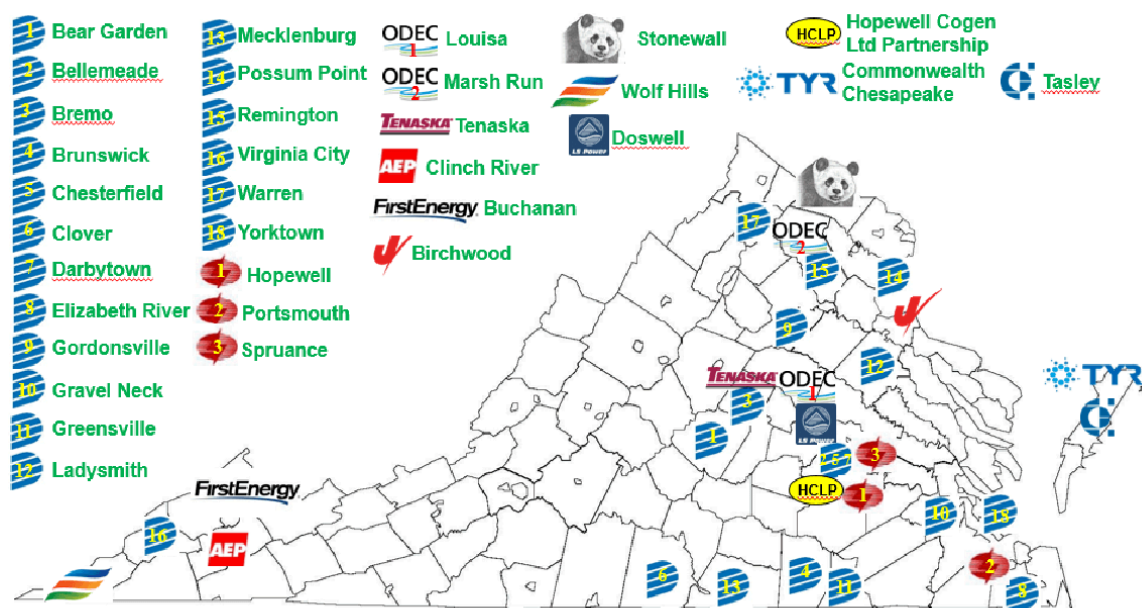
Table 1.5b: Proposed Fossil Fuel Power Plants in Virginia

Utility Name	Plant Name	County	Technology	Name-plate Capacity (MW)	Effective Month	Effective Year	Current Month	Current Year
Merck & Co Inc	Elkton	Rockingham	Natural Gas Steam Turbine	1.2	6	2020	1	2021
Merck & Co Inc	Elkton	Rockingham	Natural Gas Steam Turbine	1.2	6	2020	1	2021
C4GT, LLC	C4GT	Charles City	Natural Gas Fired Combined Cycle	1,060.0	3	2021	12	2022
Chickahominy Power LLC	Chickahominy Power LLC	Charles City	Natural Gas Fired Combined Cycle	572.1	4	2021	4	2024
Chickahominy Power LLC	Chickahominy Power LLC	Charles City	Natural Gas Fired Combined Cycle	572.1	5	2021	5	2024
Chickahominy Power LLC	Chickahominy Power LLC	Charles City	Natural Gas Fired Combined Cycle	572.1	6	2021	6	2024

Source: EIA-860 (2019), Table 3

Note: While these plants have received their permits, financial barriers have delayed their operation timelines.

Figure 1.3: Virginia Power Plants to be Covered Under RGGI



Source: Virginia Department of Environmental Quality (October 2018)

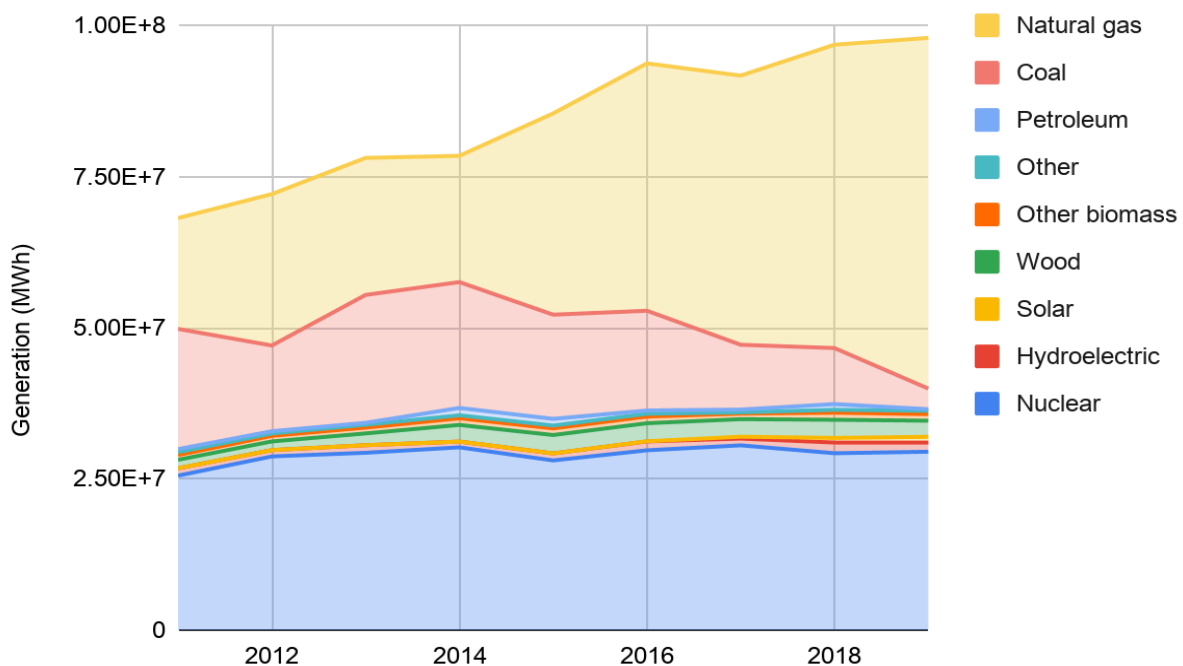
Note: Clover is half-owned by ODEC. ODEC's Wildcat Point facility is covered in Maryland.

Energy Mix

Recent Supply

- In the past decade, the predominant shift in Virginia has been from coal to natural gas. This is true for in-state electricity generation (Figure 2.1), as well as ODEC's specific portfolio (Figure 1.2).
- Outside purchases have generally dropped, while purchases of outside renewables have increased slightly ([climbing to 6%](#) of ODEC supply).
- Several GW of natural gas capacity has been added since 2017 (Table 1.5), as well as 1 GW from ODEC's Wildcat Point in Maryland. The last diesel facility came online in 2016; the rest are from 2009 or older.
 - While it is possible that more diesel could come online, natural gas plants are the most likely technology to be affected by a moratorium.

Figure 2.1: Total Electricity Generation in Virginia, 2011-2019

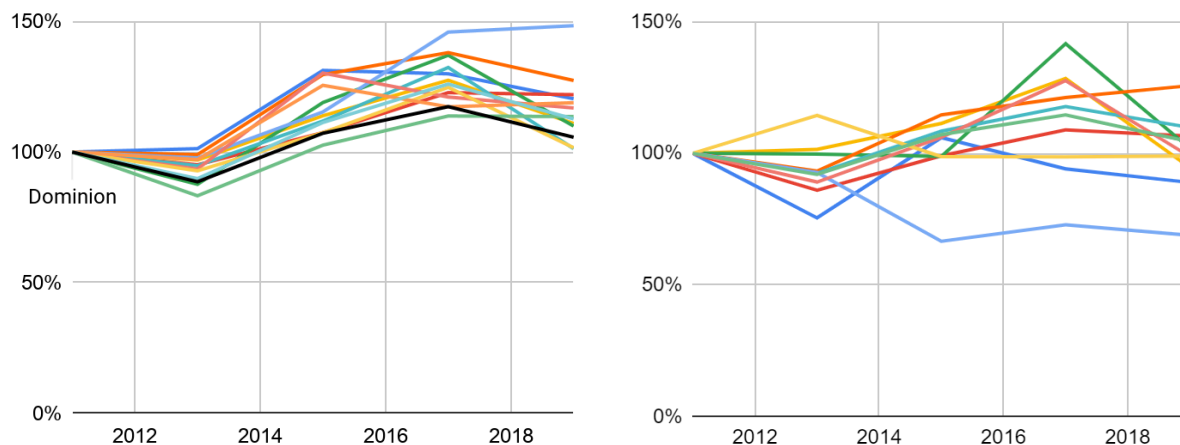


Source: EIA Table 5, Virginia state summary

Recent Demand

- Most providers have seen a slight increase in peak electricity demand, according to weather-normalized trends of winter peak demand (Figure 2.2).
 - While this metric is only one way to measure demand, winter peaks may be a useful way to consider capacity needs when outputs from renewables are lowest.
- On a percent basis, demand has increased faster in most co-ops than for Dominion. This is especially true with NOVEC, which has increased nearly 50% as more customers and data centers come online.
- Demand in munis has been more mixed, with fewer additional customers in general.

Figure 2.2: Weather-Normalized Winter Peak Demand for Virginia Co-ops and Munis



Source: EIA Form 861 (2011, 2013, 2015, 2017, 2019);

ODEC Form 10-K (2013-19) for Heating Degree Days

Maximum indicates NOVEC (left), minimum indicates City of Manassas (right)

Future Demand

- Though demand projections were not available for non-IOUs, a Greenlink [report](#) projects a stable load for APCo and gradual growth (1% per year) for Dominion. PJM [projects](#) 0.6% and 1.2% per year growth, respectively -- though such forecasts have been shown to [over-predict growth](#) due to underestimated efficiency improvements.
- Therefore, non-IOU demand may be flat in more rural areas, with increases for less rural providers like Northern Virginia Electric Cooperative (NOVEC), which may see further demand from data centers and related development.
 - If past trends continue, some co-ops may see larger increases than Dominion.
- Eventually, electricity demand may increase more broadly with electrification of transportation and buildings.

Future Generation

- Roughly 3 GW of petroleum and coal is already scheduled for retirement in 2021-23, though most is owned by Dominion (EIA Form 860 - 2019, "Generator Data").
- Now, the VCEA requires "nearly all" coal-fired plants to close by the end of 2024, [or 2025](#) for the Clover Plant that is half-owned by ODEC.
 - So, another 1.8 GW could retire by then, including 848 MW from Clover.
 - This means that ODEC will have to replace roughly 600,000 MWh of annual generation by 2025 (half of Clover's 2019 generation, via EIA Form 923 - 2019, p. 4). Clover accounted for ~12% of ODEC's generation mix in 2018 (Figure 1.2).
- If demand increases slightly, generation needs would increase from this baseline as well.
- So, by 2025, non-IOUs will likely have to replace:
 - Owned retirements - at least 12% of ODEC's 2018 generation (Clover closure)
 - Grid retirements - roughly 3-5 GW of coal and petroleum capacity in Virginia
 - The effects of grid retirements depend on PJM capacity needs. An oversupply would lessen the need for new plants to replace these retirements, while an undersupply would suggest the opposite.

Effect of a moratorium

- Without a moratorium, these future gaps could either be filled by new natural gas plants, new clean sources, or additional grid purchases.
- A moratorium on new fossil fuel power plants would prevent the natural gas option, though providers could still purchase fossil fuels from neighboring states.
- Most recent gas plants have come from third-party, merchant-owned generators (not counting ODEC's 2018 Wildcat Point facility in Maryland).
 - Since 2017, these include 1.2 GW from two operating plants, and 2.8 GW from two plants that are planned (Table 1.5).
 - Therefore, a moratorium would have the largest effect if it included these third parties, though they might simply build out-of-state.
 - A moratorium would also prevent any additional diesel plants, whose development is less likely but possible (the last new ones came online in 2016).
 - Recent plants also include 2.4 MW of an oil to gas turbine switch. This policy would also inhibit these switches (despite short-term emissions improvements), unless an exemption were made.

The above assumes a moratorium policy that simply extends the VCEA (temporary limits on VA power plants) to non-IOUs. Below are some conceivable variations (summarizing analysis by attorney David Anderson of Appalachian Voices):

- Extending temporary bans to permanent: legally possible, but not favored by most due to reliability concerns
- Extending fossil fuel bans to intrastate pipelines: legally possible, but politically tricky.
- The legality of including other infrastructure (import/export terminals, refineries, general exploration) is unclear, with potential federal complications.

- Extending the focus from banning new plants to retiring existing plants could be considered (as the VCEA does for coal). However, adequate time and stakeholder work would be necessary to avoid issues of economic transition (near power plants) and ratepayer burdens (for co-op and muni customers).
- We also note that the above focuses on electric power plants. Thousands of Virginians “purchase home heating oil, propane, gasoline and diesel fuel from a [petroleum cooperative](#)”, which relies on fossil fuel infrastructure. Future work could dig deeper into the legal, technical, economic, and political feasibility of incorporating this area.

Effect on Energy Prices

Overview

- [Report](#) that finds replacing coal with renewable energy could save customers \$8 billion each year
 - Are these Virginia co-ops mostly natural gas though? How is that different? Compare it to *operational cost* and *cost of creating new oil and natural gas infrastructure* to customers
- [Study](#) on clean energy cooperative costs in the West
- Residential electric bills are significantly lower over the 30-year period in every *zero-carbon* scenario. Compared to business as usual, the average total household savings from 2020 to 2050 range from \$1,500 under the Zero Carbon 2030 Scenario to \$3,500 under Zero Carbon 2050. - From Amber, [Advanced Energy Economy](#)

Proposed C4GT merchant plant in VA

Current cost estimates from the SCC say ratepayers will owe 5% (almost \$16 million) of construction costs, but initial cost estimates often significantly increase later. VNG plans to pay for HIP over seventy years, but its contracts with C4GT, Columbia Gas, and Dominion only last twenty years. If these parties opt not to renew their contracts, due to C4GT closing early or other reasons, ratepayers may be charged a larger share of HIP costs. The SCC Division of Utility Accounting and Finance Deputy Director said that the project should be completely paid off during the current contract with the C4GT which is 20 years.¹ Additionally, by paying back over a 70 year timeframe instead of a 20 year timeframe, lifetime costs would be around \$1.4 billion versus \$913 million. VNG states that changing the payment timeframe could make the project too expensive to build, however the SCC doubts whether the current timeframe adequately incorporates risk from Virginia's commitment to a carbon free grid in the future.²

Dominion releases Internal Resource Plan (IRP) every year. May 2020 had 4 models to include different accommodations to VCEA. All the models predict **small rate increases**.

- For example, one VCEA-compliant model predicts an average monthly rate of \$183.50 in 2030, which is "an increase of \$67.32 or 58% compared to the typical monthly bill as of May 1, 2020 and an annual increase of \$807.84"
- However, VA gov't regulatory staff reporting on the IRP finds that Dominion failed to include a least-cost model that is VCEA-compliant, thus the rates of increase could potentially be lower than reported. Sept 29, 2020

¹<https://www.virginiamercury.com/2020/04/23/as-regulators-ready-for-gas-expansion-hearings-opponents-worry-pandemic-will-have-chilling-effect-on-participation/>

²<https://www.virginiamercury.com/2020/05/19/with-new-energy-regime-only-months-away-regulators-grapple-with-gas-expansion-proposal/>

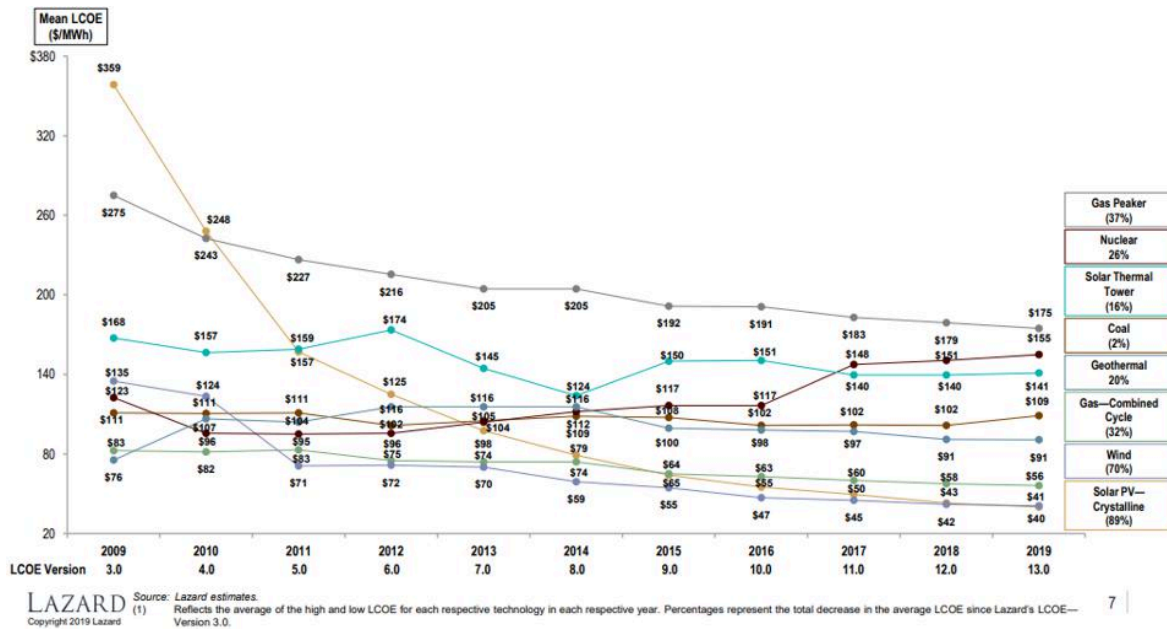
Cases of Renewable Energy Transition Effect on Electricity Retail Rates

- [California Electricity Prices Shift Due to Reliance on Renewables](#)
 - Heat wave in August - “Wholesale power prices have occasionally shot up past \$1,000 per megawatt hour — mirroring prices from the devastating 2001 energy crisis”
 - Can’t find evidence that this price spike made its way to paying customers eventually
 - Due to heavy reliance on solar, and because the heat wave effect all of the west and thus California wasn’t able to use others states’ energy
 - “the utilities could ride out the latest spikes because the higher wholesale prices are for limited periods of time and for limited amounts of energy.”
 - Costs not as significant as 2001 crisis, not affecting future rates
- [Texas’ Power Price Spike and Designing Markets for a Carbon-Free Grid](#)
 - Texas’ summer heat brings scarcity prices that encourage the construction of new projects to restore some cushion to low reserve margins and a return to low power prices.
 - ERCOT’s year-round average power prices are expected to rise over 10 percent by 2030, but only to a still-low \$33 per megawatt-hour. By contrast, August on-peak power prices are expected to fall by a nearly identical amount by 2030.
 - Texas will likely be “policy-light” in comparison to California’s policies regarding solar
 - “At majority wind and solar penetrations (i.e., more than 50 percent to 60 percent), the market need will shift from daily dispatch to monthly and seasonal availability. Energy price signals are likely to be insufficient to bring new capacity or alternative solutions to market”
 - “extreme volatility and unbankable uncertainty will compel many market operators and regulators to redesign power markets to meet the low-cost, reliable and carbon-free mandate.”
 - “In the warming world, extreme events will become both more regular and more intense and serve as a valuable benchmark in determining if power markets are able to deliver their expanded mandate of reliable, low-cost and carbon-free power.”

Costs of Renewables

- [IRENA](#) international overall recent trends in renewable power generation costs
 - From 2010-2019, solar fell 82%, onshore wind 40%, and offshore wind 29%
- [EIA](#) US projected renewable electricity generation costs 2010-2050

Figure 3.1: United States LCOE analysis by Lazard shows decreasing renewables costs.



Source: Lazard 2019

Effect of Renewables on Retail Rates

- [Prices rise while generation costs fall](#)
 - Average cost of variable renewable energy (VRE) much higher than marginal cost (because VRE doesn't require constant fuel like ff)
 - "Wholesale prices" = paid to generators, pushed down by VRE
 - "Retail prices" = paid by customer, also include the cost of grid maintenance, intermittency costs, transmission/distribution of elec ([generation only accounts for 44% of total cost](#))
 - "To handle the intermittency of VRE, system operators need to activate ramping resources more frequently to meet demand. These flexible plants are typically more expensive to operate and thus higher deployment can raise total system costs even as renewable costs decline"
- [The National Renewable Energy Lab reports](#) that a large grid system with 30 percent VRE can operate with minimal system disruption. Going beyond 30 percent, however, can present new challenges"
 - "If private capital is to fuel the renewables wave, it seems likely the sector will need to move from a more commoditized market where revenues are determined by a price per kilowatt hour to one in which revenue is generated by providing

guaranteed delivery of electricity when it is needed from sources either desired by the customer or required by regulation”

- [Effect of Renewable Portfolio Standard \(RPS\) on utility rates](#) (article locked, abstract available)
 - RPS is a costly constraint on utilities, raises rates
 - “Costs imposed on utilities affected by the RPS mandate tend to be **fixed costs** rather than variable costs.” i.e. not dependent on marginal increases in RPS requirement
- [Average retail electricity prices increase 11% 7 years after RPS](#)
 - However “the cost per metric ton of CO2 abated exceeds \$115 in all specifications and ranges up to \$530, making it least several times larger than conventional estimates of the social cost of carbon. These results do not rule out the possibility that RPS policies could dynamically reduce the cost of abatement in the future by causing improvements in renewable technology”!
 - We should take into account the social costs of carbon, as we will explore in Section 5: Health Costs.
- [Berkeley Lab](#) - 2017 literature review on Retail Rate Impacts of Renewable Electricity
- Studies on effect of renewable on electricity retail prices/premiums
 - Texas: “[there is no longer a statistically significant renewable energy premium in Texas's retail electricity pricing.](#)”
 - EU: [Impact of renewable energy regulation on retail electricity prices in the EU](#)
 - Though elec prices have risen in general, only a small portion is due to renewable energy sources: “An increase of 1% in those costs induces an average increase of only 0.023% in industrial retail prices and 0.008% in the residential retail prices.”
 - Australia: “[RET costs for households could be reduced if merit order effects were adequately passed through.](#)”
 - [Merit order effect](#): **Increasing the supply of renewable energy tends to lower the average price per unit of electricity** because wind energy and solar energy have very low marginal costs (Germany example on wiki)
- Effect of solar PV on prices
 - [Solar PV effect on ratepayers and utilities](#) (Table 3.1)
 - Effect on ratepayers and solar capacity for different PV models in Chicago, Phoenix, and Seattle
 - Batteries only effective (55% cost reduction) without net metering
 - Cost recovery for solar PV
 - [Impact of renewable PV in California penetrations on bill savings with net metering](#)
 - “High renewables penetration can lead to lower bill savings with time-varying rates.” but of uncertain value
 - **Fixed and operating costs of fossil fuel infrastructure compared to fixed and operating costs of renewables infrastructure [renewables cheaper after x years?]**

- [Utilities are transitioning to renewables as the levelized cost of electricity \(LCOE\) continues to decrease.](#)

Table 3.1: Electricity rate cost recovery for solar PV in Chicago, Phoenix, and Seattle.

Table 10

Electric rate increase required to recover utility revenue loss at various solar PV penetration rates with optimal solar PV capacity for the ratepayer (reference case shown for 0.24 \$/kW h summer on-peak price).

Location	Homes with PV (%)	Rate increase (%)	Rate price [\$/kW h] required to recoup revenue loss		
			Non-summer	Summer off-peak	Summer on-peak
Chicago	0	–	0.120	0.160	0.240
	20	16	0.139	0.186	0.278
	40	38	0.166	0.221	0.331
	60	72	0.206	0.275	0.413
	80	120	0.264	0.352	0.528
	100	192	0.350	0.497	0.701
Phoenix	0	–	0.120	0.160	0.240
	20	24	0.149	0.198	0.298
	40	63	0.196	0.261	0.391
	60	135	0.282	0.376	0.564
	80	268	0.442	0.589	0.883
	100	466	0.680	0.906	1.358
Seattle	0	–	0.120	0.160	0.240
	20	8	0.130	0.173	0.259
	40	19	0.143	0.190	0.286
	60	32	0.158	0.211	0.317
	80	47	0.176	0.235	0.353
	100	66	0.199	0.266	0.398

Source: Janko et al. (2016)

- [Utility Financial Transition Impacts](#)
 - [“Steel-to-Fuel”](#) reduces risk and creates opportunities for equity shareholders by moving capital from ff to renewable

Economic Benefits and Job Creation

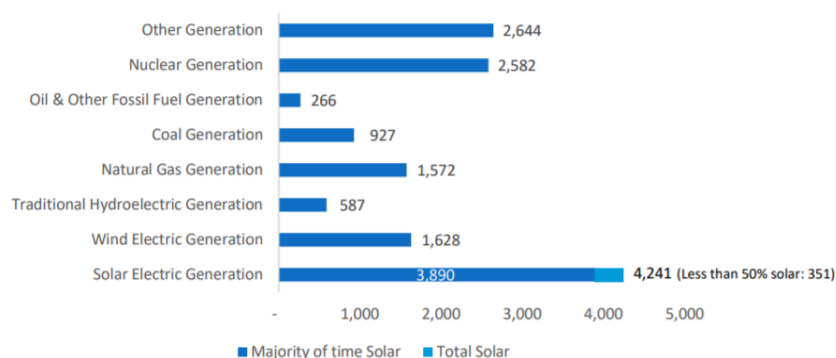
Employment sectors

- A few co-ops have local renewables like community solar. ODEC has 2 natural gas, 1 coal, 3 solar, 1 nuclear, 1 LFG plant in VA.
- Many munis have local petroleum plants, a few have hydro
- [Virginia generally](#), 2019
 - Electric Power Generation employs 14,446 workers in Virginia, 1.6 percent of the national total and adding 550 jobs over the past year (4.0 percent). *Solar makes up the largest segment of employment related to Electric Power Generation*, with 4,241 jobs (up 1.0 percent), followed by traditional fossil fuel generation at 2,765 jobs (up 3.7 percent).
 - Construction is the largest industry sector in Electric Power Generation, with 31.6 percent of jobs. Professional and business services are next with 29.6 percent.
 - Fuels employs 13,957 workers in Virginia, 1.2 percent of the national total, up 7.7 percent over the past year. Petroleum and other fossil fuels makes up the largest segment of employment related to Fuels
 - Professional and business services jobs represent 33.5 percent of Fuels jobs in Virginia.
 - Transmission, Distribution, and Storage employs 25,352 workers in Virginia, 1.9 percent of the national total, up 3.7 percent or 911 jobs since the 2018 report.
 - The 78,670 Energy Efficiency jobs in Virginia represent 3.4 percent of all U.S. Energy Efficiency jobs, adding 2,049 jobs (2.7 percent) since last year. The largest number of these employees work in ENERGY STAR and efficient lighting firms, followed by traditional HVAC.
 - Energy Efficiency employment is primarily found in the construction industry.

Figure 4.1: Electric Power Generation Employment by Detailed Technology Application

Figure VA-2.

Electric Power Generation Employment by Detailed Technology Application

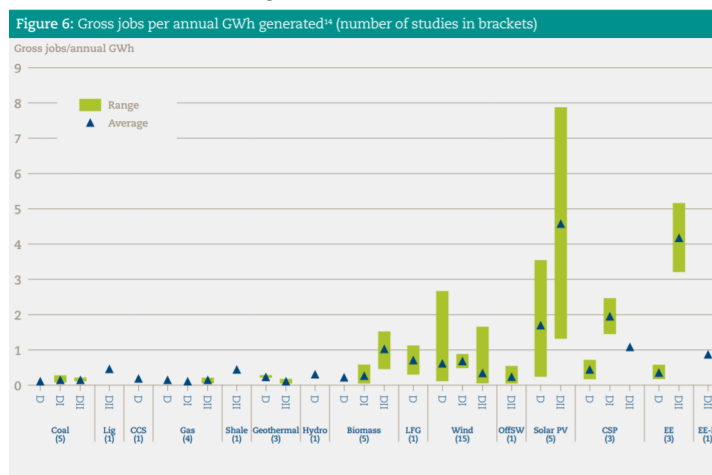


Source: US Energy and Employment Report, 2019

Job Creation

- An [article in Science](#) found that, using a generator-level model, “in the United States, a 2035 electricity decarbonization deadline, as proposed by President-elect Biden and the 2020 Democratic party platform (6, 7), would strand only about 15% of fossil capacity-years and 20% of job-years, which is unusually low from a global perspective”
- Another [study](#) finds that “2.65 full-time-equivalent (FTE) jobs are created from \$1 million spending in fossil fuels, while that same amount of spending would create 7.49 or 7.72 FTE jobs in renewables or energy efficiency. Thus each \$1 million shifted from brown to green energy will create a net increase of 5 jobs.”
- A 2019 report from [Advanced Energy Economy](#) finds that transitioning to a zero-carbon grid in Virginia by 2030, 2040, or 2050 would lead to benefits in GDP, income, and job creation.
 - All zero-carbon scenarios led to net job growth from new energy efficiency measures and new renewable and battery storage resources. On average, Zero Carbon Scenario job creation exceeds business as usual by an average of 7,000 to 13,000 jobs per year.
 - Residential electric bills are significantly lower over the 30-year period in every zero-carbon scenario. Compared to business as usual, the average total household savings from 2020 to 2050 range from \$1,500 under the Zero Carbon 2030 Scenario to \$3,500 under Zero Carbon 2050.
- The [Bureau of Labor Statistics](#) forecasts that solar and wind technicians will be among the top three fastest growing jobs from now until 2029, at 61% and 51% growth respectively. Comparatively, oil and gas laborers and operators will experience a 25-30% growth rate.
- According to a [study by the UKERC](#), Solar PV, CSP, and Energy Efficiency create more jobs per GWh generated compared to oil and gas.

Figure 4.2: Gross jobs per annual GWh generated. *Direct (D), indirect (DI) and induced (DII).*



Source: UKERC Technology & Policy Assessment Function, 2014

In the event of renewables replacing fossil fuel infrastructure in Virginia, a lack of jobs is not as much of a concern as the ability to effectively transition and train workers in renewable energy jobs. **Articles on the time frame for this transition?**

Effect on Health

Compiled Research:

- [This from EPA](#): Estimating (Monetized) Health Benefits per Kilowatt-hour of Energy Efficiency and Renewable Energy
 - Based on 2017 electricity generation data and emissions, population, baseline mortality incidence rate, and income growth projections
 - To use: multiply the benefits per kWh values by the number of kilowatt-hours saved from EE or generated from RE to estimate potential health benefits from projects in dollars saved
 - Virginia is split between Southeast and Mid-Atlantic Regions
 - Example: For Mid-Atlantic Regions, low estimate of 3.67 cents per kw-hr for solar projects
 - [Public Health Benefits per kWh of Energy Efficiency and Renewable Energy in the United States: A Technical Report | US EPA](#): explains how they determined these benefits
 - [CO-Benefits Risk Assessment Health Impacts Screening and Mapping Tool](#): estimates health and economic benefits of clean energy policies, specifically focused on costs of air pollution. Can
 - Explore how changes in air pollution from clean energy policies and programs, including energy efficiency and renewable energy, can affect human health at the county, state, regional, or national levels.
 - Estimate the economic value of the health benefits associated with clean energy policies and programs to compare against program costs.
 - Map and visually represent the air quality, human health, and health-related economic benefits from reductions in emissions
 - [how to use](#)
 - Need to input a percentage when a policy is expected to reduce emissions or use of an energy source by a specific proportion
- New York Dept. of Public Services: [White Paper on Benefit-Cost Analysis in the Reforming Energy Vision Proceeding \(July 2015\)](#)
- [Climate and health benefits of increasing renewable energy deployment in the United States](#)
 - develops a ten-region model to assess the magnitude of health and climate benefits across the US. Then uses this model to assess the benefits of deploying varying capacities of wind, utility-scale solar photovoltaics (PV), and rooftop solar PV in different regions in the US
- <https://ideas.repec.org/a/eee/renene/v68y2014icp51-57.html>
 - “this paper provides an economic valuation of RES-E promoted by the Spanish feed-in system. First, supported RES-E is evaluated in terms of CO2 emissions avoided when they are introduced in the Spanish electricity market instead of other potential polluting energy sources. And secondly, these positive

environmental externalities of supported RES-E are compared with the funds they received from the Spanish feed-in system, in order to estimate the economic balance of this support system. “

- [Health Care Sector Reasons to Oppose Excessive Pipeline Development](#)
 - Uses Healthcare Energy Impact Calculator, www.eichealth.org
- **Environmental Justice issues with proposed C4GT and Chickahominy plants:** The Gidley compressor station would be put by 6500 people in a one mile radius with that population being 65% POC and 31% low-income. Additionally, Charles City County which is a majority minority community will receive new gas pipelines, as well as the construction of two large gas plants. The C4GT plant, originally cited as the reason for the Header Improvement Project, would be constructed within 1.1 miles of another gas plant, the Chickahominy Plant. Charles City County also already suffers from proximity to a mega-landfill that has a history of leaching.³⁴

Potential Complementary Policies

- Current Federal Renewable Incentives/Policies
 - Financing through NRECA, [NRUCFC](#) (NGO), and Department of Agriculture
 - USDA REAP grants apply to co-ops
 - According to the US Department of Agriculture (USDA), the [Rural Energy for America Program \("REAP"\)](#), enacted in 2008, "provides guaranteed loan financing and grant funding to agricultural producers and rural small businesses for renewable energy systems or to make energy efficiency improvements."
 - [Renewable Electricity Production Tax Credit \(PTC\)](#)
 - Per kilowatt-hour (kWh) tax credit for producers of renewable energy, which ranges from 1.2 to 2.3 cents/kWh.
 - [NRECA says](#) "Although not-for-profit co-ops are not typically eligible for tax benefits, they often seek a "taxable partner" for solar and wind projects, either through a power-purchase-agreement or through a shared ownership model, such as a tax-equity flip or a tax-lease-buyback project."
 - [Solar Investment Tax Credit \(ITC\)](#) is a 26% tax credit for residential, commercial, and community solar arrays, which has spurred 52% average annual growth of solar since it was enacted in 2006. However, this tax credit, which was passed under the Energy Policy Act in 2005, will expire in 2022 unless it is renewed by congress.
 - Net metering is also available to electrical cooperatives.
 - EPA [Clean Energy Incentive Program](#) will provide Emission Rate Credits (ERCs)
 - [Virginia's Renewable Portfolio Standard \(RPS\)](#) is not compulsory but requires IOU's to report efforts to meet its goals. The RPS generates renewable energy credits (RECs).
 - Goal: 100% renewables by 2045 for Phase II utilities and 2050 for Phase I utilities
 - [Only applies to IOU's](#), though many other states do include municipal, cooperative, and retail utilities.
- Barriers to energy transitions
 - Financing Issues
 - Many co-ops are in debt. This [article](#) states that "the federal government is on the hook for about \$8.4 billion of loan guarantees on existing coal assets for co-ops... rural communities could better pursue a clean energy future if current debt on existing coal plant infrastructure could be eliminated in exchange for a requirement to invest in clean energy and energy efficiency."
 - Contracts
 - Many co-ops are stuck in long term fossil fuel agreements. Some do say they support current fossil fuel infrastructure during the transition due to

grid reliability, but if done in a way that allows flexibility, co-ops have been shown to easily transition. [News article for context](#)

- **Suggestions for VA co-ops and munis**

- Munis/co-ops may argue that RPS may be too stringent.
- [Grid modernization](#)
 - Energy Efficiency Resource Standards (EERS) established through the VCEA do not apply to co-ops/munis
 - Co-ops and munis may naturally want to modernize the grid to save money. For low population areas, “smart meters decrease costs by eliminating in-person meter-reading, while offering improved service, reduced outages and quicker recovery times.”
 - Decoupling revenue from sales to reduce pressure to sell more electricity (as done by the LA Department of Water and Power) can facilitate grid modernization from co-ops/munis.
- Clean power skills training (helping coops/munis transition)
- Incentives for clean purchases such as tax credits